

IndiaKart E-Commerce Analytics

Comprehensive Management & Operations Final Project Report (Phases 1–4)

Prepared By: Netra Bolishetti Scope: Data Quality, EDA, KPIs, Power BI & Strategy Period: 24 Months Operations

Project Context & Scope

IndiaKart analytics covers 8 interconnected relational datasets comprising ~216,200 records over 24 months of business operations. The analysis delivers actionable strategic guidance on category revenue, order cancellations, customer value, seasonality, geographic performance, returns, payments, and warehouse inventory.

1. DATASET ARCHITECTURE

Table Name	Record Count	Core Business Purpose
orders	50,000	Central order header data & transaction lifecycle timestamps
order_items	100,000	Product-level transaction details & line-item calculations
payments	50,000	Payment transaction records, gateways, and methods
customers	10,000	Customer profiles, demographics, and behavioral segments
returns	10,000	Return request logs, authorization, and reason classification
products	1,000	Master product catalog, category definitions, and pricing
inventory	1,000	Warehouse stock balances, reorder thresholds, and safety levels
suppliers	200	Vendor directory, fulfillment metrics, and performance ratings

2. PHASE 1: DATA CLEANING & QUALITY AUDIT

All 8 datasets underwent rigorous profiling for data types, missing values, duplicates, outliers, and relational integrity. Key validation outcomes include:

- Zero Unmatched Records:** Referentials passed perfectly across `order_items` → `orders`, `order_items` → `products`, `returns` → `orders`, and `inventory` → `products`.
- Missing Data Integrity:** Unfulfilled orders legitimately had null delivery dates; refund fields in 50,000 payment rows indicated zero recorded direct refunds in gateway logs.
- Clean Customer & Product Catalogs:** Master entities (10,000 customers, 1,000 products) contained 0 duplicates and 0 missing values.

3. PHASE 2: EXPLORATORY DATA ANALYSIS KEY HIGHLIGHTS

Analysis Vector	Core Finding / Metric	Strategic Implication
Peak Monthly Volume	Jan 2025 (2,239 orders)	Strong Q1 seasonal demand momentum
Category Revenue Share	Electronics: ₹1.64B (57.41%)	High revenue concentration risk
Order Status Breakdown	65.0% Delivered 11.99% Cancelled	High cancellation loss stream
Top Market State	Uttar Pradesh	Primary order volume geographic driver

4. PHASE 3: BUSINESS KPI SCORECARD

GROSS MERCHANDISE VALUE ₹3.15B 50,000 Total Orders	NET REVENUE ₹2.05B Delivered Orders Only	AVERAGE ORDER VALUE ₹63,197.59 Delivered Orders Baseline	RETURN RATE 30.77% Target: < 8.00%
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Key Performance Indicator	Calculated Value	Benchmark / Target	Performance Status
Cancellation Rate	11.99%	< 10.00%	Above Target
Return Rate	30.77%	< 8.00%	Critical Risk
Payment Failure Rate	3.50%	< 2.00%	Action Needed
Inventory Fill Rate	92.20%	> 95.00%	Underperforming
Top Category Share (Electronics)	57.41%	< 60.00%	Within Bounds
CLV – Premium Segment	₹532,735.00	1.47x Regular CLV	High Value Target
CLV – Regular Segment	₹361,870.72	Baseline Segment	Core Volume Base
MoM GMV Volatility	+19.61% / -23.74%	Stable Growth	High Seasonality

5. CUSTOMER SEGMENT AOV BREAKDOWN

Average Order Value (AOV) by Segment - Budget Segment: ₹64,124.56 - New Customers: ₹63,381.58 - Regular Segment: ₹63,123.55 - Premium Segment: ₹61,876.90 - Inactive Segment: ₹57,120.14	Segment Observations While Budget and New customer segments demonstrate slightly higher single-transaction AOV, Premium customers hold dramatically higher Lifetime Value (₹532.7k vs ₹361.8k) due to repeat order frequency.
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6. PHASE 4: POWER BI DASHBOARD STRUCTURE

Dashboard Page	Core Visuals & Features	Primary Slicers & Interactivity
Page 1: Revenue Overview	Monthly GMV trend, order volume, total GMV, Net Revenue, AOV card, cancellation metric	Date range slicer, order status filter
Page 2: Category Performance	Revenue share by category, top 10 products, category-wise return rate matrix	Category slicer, product group dropdown
Page 3: Customer Insights	Segment breakdown donut chart, top 10 states volume map, monthly acquisition chart	Customer segment slicer, region filter
Page 4: Operations	Order status conversion funnel, payment failure analysis, return reason breakdown	Order status slicer, payment gateway selector
Page 5: Inventory Alert	Warehouse stock summary, low-stock threshold table, stock-out risk cards	Warehouse location slicer, supplier filter

7. EXECUTIVE MANAGEMENT SUMMARY

IndiaKart achieved **₹3.15 Billion in GMV** across 50,000 orders over the 24-month analysis period, generating **₹2.05 Billion in net revenue** from delivered orders with an Average Order Value (AOV) of ₹63,197.59. Despite strong sales volume, critical operational friction impacts net profitability.

Top Operational Risks

- **Excessive Return Rate (30.77%):** Severely damages margins via reverse logistics, processing costs, and inventory ties.
- **High Cancellation Rate (11.99%):** Results in lost revenue opportunity and unnecessary fulfillment overhead.
- **Payment Failures (3.50%):** Exceeds 2% benchmark, leading to checkout friction and abandoned carts.
- **Inventory Stock-outs (92.2% Fill):** Below 95% target, risking unfulfilled demand during peak sales.

Key Strategic Opportunities

- **Category Diversification:** Reduce reliance on Electronics (57.41% share) by expanding Lifestyle and Home sectors.
- **Premium Segment Retention:** Capitalize on Premium CLV (₹532,735) through targeted loyalty programs.
- **Sizing & Quality Assurance:** Mitigate primary return cause ("Size Issue", 1,196 cases) via enhanced catalog charts.
- **Payment Gateway Optimization:** Streamline UPI and card checkout pathways to reduce failures.

8. STRATEGIC RECOMMENDATIONS ROADMAP

Action Area	Recommended Operational Focus	Expected Impact
1. Returns Reduction	Audit top returned SKUs; update sizing guides; introduce pre-shipment quality verification.	High Margin Gain
2. Cancellation Prevention	Analyze cancellation timing; send automated SMS/WhatsApp order updates; shorten dispatch cycles.	+2–4% Net Rev
3. Payment Reliability	Implement smart payment routing between multi-bank gateways to handle UPI spikes.	Cart Recovery
4. Inventory Management	Re-calibrate safety stock levels for fast-moving SKUs; establish automated supplier reorder triggers.	95%+ Fill Rate
5. Customer Retention	Establish VIP concierge & early-access promotions for Premium and Regular repeat segments.	CLV Expansion

9. DELIVERABLES MANIFEST & SUBMISSION PACKAGE

Final Project Repository Structure

- Phase1_DataCleaning_Netra.ipynb – Cleaning code, missing data profiling & relational checks.
- Phase2_EDA_Netra.ipynb – Exploratory analysis code & 10 primary charts.
- EDA_Charts/ – Directory containing saved high-resolution PNG chart outputs.
- Phase3_KPIs_Netra.ipynb – Executable code for 10 core business KPI calculations.
- Phase4_Dashboard_Netra.pbix – Interactive 5-page Power BI dashboard file.
- Phase4_Report_Netra.pdf – Executive summary report for management review.